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MTH501 Linear Algebra

Mid Term Examination - November 2004

Time Allowed: 90 Minutes

Instructions

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions. Marks are written adjacent to each question.
2. Do not ask any questions about the contents of this examination from anyone.
 - a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.
 - b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an 'F' grade in this course.**

Total Marks: 45
Questions: 11

Total

Question No. 1	Marks : 1
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Consider the system of Linear equations

$$x_1 + 4x_2 - 5x_3 = 0$$

$$2x_1 - x_2 + 8x_3 = 9$$

$$x_1 + x_2 - 8x_3 = 2$$

Then determinant of the Matrix of Coefficients of the above system

- is (a) 91
(b) 123
(c) 141
(d) **81**

Question No. 2**Marks : 2**

Consider the matrix $A = \begin{bmatrix} 12 & x+y \\ 23 & 4 \\ 23 & x-y \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 & x-5y \\ 2 & 1 & 1 \\ 0 & 2 & 3x \end{bmatrix}$ such that

$$AB = \begin{bmatrix} 2020 \\ 445 \\ 2512 \end{bmatrix}$$

Then the values of x and y are

- (a) $x = -2$ and $y = 4$
- (b) $x = 2$ and $y = -6$
- (c) $x = 0$ and $y = 0$
- (d) $x = 2$ and $y = -4$**

Question No. 3**Marks : 1**

Which statement about the set S is false where $S = \{(1, 1, 3), (2, 3, 7), (2, 2, 6)\}$

- (a) The set S contain an element which is solution of the equation $5x - y - z =$
- (b) The Set S is linearly independent.**
- (c) The set S contain two elements which are multiple of each
- (d) The set S is linearly dependent.

Question No. 4**Marks : 2**

Consider a linear transformation T such that $T \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$ Then $T \begin{bmatrix} 5 \\ 25 \\ 15 \end{bmatrix} =$

- (a) $\begin{bmatrix} 9 \\ 18 \end{bmatrix}$
- (b) $\begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$
- (c) $\begin{bmatrix} 30 \\ 45 \end{bmatrix}$**
- (d) $\begin{bmatrix} 12 \\ 9 \end{bmatrix}$

Question No. 5**Marks : 1**

How many subspaces R^2 have?

- (a) only two: $\{0\}$ and R^2
- (b) Only four: $\{0\}$ x-axis and y-axis and R^2
- (c) Infinitely many.**
- (d) None of the above.

Question No. 6**Marks : 8**

An oil refinery produces low-sulfur and high-sulfur fuel. Each ton of low-sulfur requires 5 minutes in the blending plant and 4 minutes in the refining plant; each ton of high sulfur fuel requires 4 minutes in the blending plant and 2 minutes in the refining plant. If the blending plant is available for 3 hours and the refining plant is available for 2 hours, how many tons of each type of fuel should be manufactured so that the plants are fully utilized?

Solution:

The data given in the question can be formed into system of linear equations as

Low-sulfur High-sulfur

Blending plant 5 4

Refining Plant 4 2

As we are given in the question that we have blending plant and refining plant available for 3 and 2 hours respectively. Let x tons of low-sulfur and y tons of high sulfur be the amount should be manufactured so that plants are fully utilized.

Therefore from the above data we must have the system,

$$5x + 4y = 180$$

$$4x + 2y = 120$$

Augmented matrix for the above

$$\text{system} \begin{bmatrix} 5 & 4 & 180 \\ 4 & 2 & 120 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 60 \\ 4 & 2 & 120 \end{bmatrix} \text{ by } R_1 - R_2$$

$$\text{is } \sim \begin{bmatrix} 1 & 2 & 60 \\ 0 & -6 & -120 \end{bmatrix}$$

So we have $x + 2y = 60$ and $-6y = -120$, thus we have $y = 20$ tons and $x = 20$ tons are the required manufactured tons of each low-sulfur and high-sulfur so that we can utilize both plants for the given time.

Question No. 7**Marks : 5**

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 4 & 1 & 1 \\ 7 & 0 & 2 \end{bmatrix}$$

Consider the matrix A then find the entry a_{33} of the matrix A^2 without calculating the matrix A^2 .

Solution:

$$\text{Required entry } a_{33} \text{ of } A^2 = \begin{bmatrix} \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} \\ 7 & 0 & 2 \end{bmatrix} \begin{bmatrix} \text{---} & \text{---} & 2 \\ \text{---} & \text{---} & 1 \\ \text{---} & \text{---} & 2 \end{bmatrix} \text{ marks:3} = \begin{bmatrix} \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} \\ 14+0+4 & \text{---} & \text{---} \end{bmatrix} = \begin{bmatrix} \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & 18 \end{bmatrix}$$

Question No. 8**Marks : 8**

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 4 & 1 & 1 \\ 7 & 0 & 2 \end{bmatrix}$$

Find LU decomposition for the matrix

Solution:

$$\begin{array}{l}
 U \qquad \qquad \qquad L \\
 \left[\begin{array}{ccc} 1 & \frac{1}{3} & \frac{2}{3} \\ 4 & 1 & 1 \\ 7 & 0 & 2 \end{array} \right] \text{multiplier } \frac{1}{3} \left[\begin{array}{ccc} 3 & 0 & 0 \\ * & * & 0 \\ * & * & * \end{array} \right] \\
 \left[\begin{array}{ccc} 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & -\frac{1}{3} & -\frac{5}{3} \\ 0 & -\frac{7}{3} & -\frac{8}{3} \end{array} \right] R_2 - 4R_1, R_3 - 7R_1 \left[\begin{array}{ccc} 3 & 0 & 0 \\ 4 & * & 0 \\ 7 & * & * \end{array} \right] \\
 \left[\begin{array}{ccc} 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & 1 & 5 \\ 0 & -\frac{7}{3} & -\frac{8}{3} \end{array} \right] \text{multiplier } -3 \left[\begin{array}{ccc} 3 & 0 & 0 \\ 4 & -\frac{1}{3} & 0 \\ 7 & * & * \end{array} \right] \\
 \left[\begin{array}{ccc} 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & 1 & 5 \\ 0 & 0 & 9 \end{array} \right] R_2 + \frac{7}{3}R_2 \left[\begin{array}{ccc} 3 & 0 & 0 \\ 4 & -\frac{1}{3} & 0 \\ 7 & -\frac{7}{3} & * \end{array} \right] \\
 \left[\begin{array}{ccc} 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{array} \right] \text{multiplier } -\frac{1}{9} \left[\begin{array}{ccc} 3 & 0 & 0 \\ 4 & -\frac{1}{3} & 0 \\ 7 & -\frac{7}{3} & 9 \end{array} \right] .
 \end{array}$$

The required LU factorization is

$$L = \begin{bmatrix} 3 & 0 & 0 \\ 4 & -\frac{1}{3} & 0 \\ 7 & -\frac{7}{3} & 9 \end{bmatrix} \quad U = \begin{bmatrix} 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$$

Question No. 9

Marks : 8

Write $v = \begin{bmatrix} 4 \\ 9 \\ 2 \end{bmatrix}$ as linear combination of $u_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, $u_2 = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$ and $u_3 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$. (That is you have to find out the constants such that $v = c_1 u_1 + c_2 u_2 + c_3 u_3$)

Solution:

We will try to find out the constants such that we can write the augmented matrix of the system of linear equations correspond to the linear combination as

$$\begin{bmatrix} 1 & 1 & 1 & 4 \\ 2 & 4 & -3 & -9 \\ -1 & 2 & 2 & 2 \end{bmatrix} \quad \text{And echelon form of that matrix is } \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 2 & 6 \\ 0 & 0 & 1 & 3 \end{bmatrix} \Rightarrow x_3 = 3, x_2 = -1, x_1 = 2$$

Question No. 10

Marks

: 5

Show that the mapping $M: R^3 \rightarrow R^3$ defined by $M \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ is a linear transformation. Also write the matrix of that linear transformation.

Solution:

We will prove that the mapping satisfies the

$$(i) \quad M(u + v) = M(u) + M(v) \quad (ii) \quad M(cu) = cM(u)$$

(i) $M \begin{bmatrix} a \\ b \\ c+f \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c+f \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} d \\ e \\ f \end{bmatrix} \right\}$ Now as we know that matrices are distributed over addition that is if A, B and C are matrices then $A(B+C) = AB + BC$, so we can write,

$$M \begin{bmatrix} a \\ b \\ c+f \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} d \\ e \\ f \end{bmatrix} \right\} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} d \\ e \\ f \end{bmatrix} \text{ also by}$$

definition we have $M \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ and $M \begin{bmatrix} d \\ e \\ f \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} d \\ e \\ f \end{bmatrix}$. Hence we have,

$$M \begin{bmatrix} a \\ b \\ c+f \end{bmatrix} = Mb + Me + \begin{bmatrix} d \\ e \\ f \end{bmatrix}$$

$$M \begin{bmatrix} cd \\ ce \\ cf \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} cd \\ ce \\ cf \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \left\{ c \begin{bmatrix} d \\ e \\ f \end{bmatrix} \right\} = c \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} d \\ e \\ f \end{bmatrix}$$

As we have $A(cB) = cAB$ where c is a real number.

$$(ii) \quad Mce \begin{bmatrix} cd \\ ce \\ cf \end{bmatrix} = cMe \begin{bmatrix} d \\ e \\ f \end{bmatrix}$$

Hence M is a linear transformation and matrix of transformation $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

Question No. 11 **Marks** : 5

Solve the system $3x + y = 12$ by using Jacobi's Method starting $x^0 = (0, 0, 0)$ and only three iterations.

$$2x - 4y = -6$$

Solution:

Iteration 1
(12/3, 3/2)
iteration 2
(3.5, 3.5)
iteration 3

(2.833, 3.25)